

**Service Information System**

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Product: BACKHOE LOADER

Model: 416F2 BACKHOE LOADER LYA

Configuration: 416F2 Center Pivot Backhoe Loader LYA00001-UP (MACHINE) POWERED BY C4.4 Engine

Operation and Maintenance Manual**416F2, 420F2 and 430F2 Backhoe Loaders****Media Number -SEBU9212-05****Publication Date -01/02/2015****Date Updated -26/02/2016**

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Lubricant Viscosities - Fluids Recommendations

SMCS - 1000; 7000

General Information for Lubricants

When you are operating the machine in temperatures below -20°C (-4°F), refer to Special Publication, SEBU5898, "Cold Weather Recommendations". This publication is available from your Cat dealer.

For cold-weather applications where transmission oil SAE 0W-20 is recommended, Cat TDTO Cold Weather is recommended.

Refer to the "Lubricant Information" section in the latest revision of the Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations" for a list of Cat engine oils and for detailed information. This manual may be found on the Web at Safety.Cat.com.

The footnotes are a key part of the tables. Read ALL footnotes that pertain to the machine compartment in question.

Selecting the Viscosity

In order to select the proper oil for each machine compartment, refer to the "Lubricant Viscosity for Ambient Temperature" table. Use the oil type AND oil viscosity for the specific compartment at the proper ambient temperature.

The proper lubricant viscosity grade is determined by the minimum outside temperature when the machine is started. The proper lubricant viscosity grade is also determined by the maximum outside temperature while the machine is operated. Use the column on the table that is designated "Min" in order to determine the lubricant viscosity grade that is required when you start a cold machine and when you operate a cold machine. Use the column on the table that is designated "Max" to select the lubricant viscosity grade when you operate the machine at the highest temperature that is anticipated.

When you start the machine, use the oil with the highest lubricant viscosity that is allowed for the temperature.

Machines that are continuously operated should use the oils with a higher viscosity in the final drives and differentials in order to maintain the highest possible oil film thickness. Refer to “General Information for Lubricants” article, “Lubricant Viscosities” tables, and any associated footnotes. Consult your Cat dealer if additional information is needed.

NOTICE

Not following the recommendations found in this manual can lead to reduced performance and compartment failure.

Engine Oil

Cat oils have been developed and tested in order to provide the full performance and life that has been designed and built into Cat engines.

Cat DEO-ULS multigrade and Cat DEO multigrade oils are formulated with the correct amounts of detergents, dispersants, and alkalinity in order to provide superior performance in Cat diesel engines where recommended for use.

Table 1

Lubricant Viscosities for Ambient Temperatures						
Compartment or System	Oil Type and Performance Requirements	Oil Viscosities	°C		°F	
			Min	Max	Min	Max
Engine Crankcase for all Machines	Cat DEO-ULS Cold Weather	SAE 0W-40	−40	40	−40	104
	Cat DEO-ULS Cat DEO	SAE 10W-30	−18	40	0	104
	Cat DEO-ULS Cat DEO	SAE 15W-40	−9.5	50	15	122

When fuels of sulfur level 0.1 percent (1000 ppm) or higher are used, Cat DEO-ULS may be used if an oil analysis program is followed. Base the oil change interval on the oil analysis.

Other Oil Applications

Refer to the “Lubricant Information” section in the latest revision of the Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations" for a list of Cat engine oils and for detailed information. This manual may be found on the Web at Safety.Cat.com.

Cat HYDO Advanced fluids are the preferred oils for use in Cat machines hydraulic systems.

Cat HYDO Advanced fluids have a 50% increase in the standard oil drain interval for machine hydraulic systems (3000 hours versus 2000 hours) over second and third choice oils when you follow the maintenance interval schedule for oil filter changes and for oil sampling that is stated in the Operation and Maintenance Manual for your particular machine. 6000 hour oil drain intervals are possible when using S·O·S Services oil analysis. Consult your Cat dealer for details. When switching to Cat HYDO Advanced fluids, cross contamination with the previous oil should be kept to less than 10%.

Note: For machines equipped with hydraulic hammers, do not use viscosity grades SAE 0W or SAE 5W oils. Refer to the “Special Applications” section in this article.

Table 2

Backhoe Loaders Lubricant Viscosities for Ambient Temperatures						
Compartment or System	Oil Type and Performance Requirements	Oil Viscosities	°C		°F	
			Min	Max	Min	Max
Direct Drive Transmissions Powershift Transmissions	Cat TDTO Cold Weather	SAE 0W-20	-40	35	-40	95
	Cat TDTO	SAE 10W	-20	35	-4	95
		SAE 30	25	50	77	122
	Cat TDTO-TMS	Multi-Grade	10	50	50	122
Front All Wheel Drive Axle and Final Drives⁽¹⁾	Cat TDTO	SAE 0W-20	-40	10	-40	50
		SAE 0W-30	-40	20	-40	68
		SAE 5W-30	-30	20	-22	68
		SAE 10W	-20	10	-4	50
		SAE 30	All temperature ranges			
		SAE 50	10	50	50	122
Hydraulic Systems	Cat HYDO Advanced 10 Cat Bio HYDO Advanced Cat Cold Weather	SAE 0W-20	-40	50	-40	122
		SAE 0W-30	-40	50	-40	122

	TDTO Cat BF-2 Cat TO-4	SAE 10W	-20	50	32	122
		Bio HYDO Advanced	-40	50	-4	122

⁽¹⁾ Refer to table 3.

Backhoe Loader Rear Axles

These recommendations are for backhoe loaders with a fixed rear axle (2 wheel steering). For backhoe loaders with All Wheel Steering (AWS) rear axles, refer to the lubricant recommendations in the machine Operation and Maintenance Manual.

Additional 197-0017 may be used in order to reduce brake noise.

Do not use Cat MTO or commercial M2C134-D specification oil with the 230-4017 brake disks. Do not use Cat MTO or commercial M2C134-D specification oil in any F Series Backhoe Loader rear axle.

Table 3

Lubricant Viscosities for Ambient Temperatures for Backhoe Loader Rear Axles							
Brake Disk Part Number	197-0017 Additive Volume	Oil Type and Performance Requirements for Rear Axle	Oil Viscosity Grade	°C		°F	
				Min	Max	Min	Max
133-7234	1L (1.1qt) ⁽¹⁾	Cat MTO or M2C134-D ⁽²⁾	SAE 30	-25	40	-13	104
133-7234	1L (1.1qt) ⁽¹⁾	Cat TDTO 30 or Cat TO -4 30					
230-4017	150mL (5.1oz) ⁽³⁾	Cat TDTO 30 or Cat TO -4 30					
238-5291	500 mL (17.0 oz) ⁽⁴⁾	Cat TDTO 30 or Cat TO -4 30					
288-7303	200mL (6.8oz) ⁽⁵⁾	Cat TDTO 30 or Cat TO -4 30					

⁽¹⁾ The maximum amount of 197-0017 for this brake is 2 L (2.1 qt).

⁽²⁾ For machines with (S/N: LBS) and (S/N: LBF) this is the only oil recommended for the front and rear axle.

⁽³⁾ The maximum amount of 197-0017 for this brake is 300 mL (10.2 oz).

⁽⁴⁾ The maximum amount of 197-0017 for this brake is 550 mL (18.7 oz).

⁽⁵⁾ The maximum amount of 197-0017 for this brake is 250 mL (8.5 oz).

Grease Applications

In order to use a non-Cat grease, the supplier must certify that the lubricant is compatible with Cat grease.

Each pin joint should be flushed with the new grease. Ensure that all old grease is removed. Failure to meet this requirement may lead to failure of a pin joint.

Table 4

Type of Caterpillar Grease by Machine Category									
Vehicle	Application Point	Typical Load and Speed	Load Factor	Ambient Temperature Range				NLGI Grade	Grease Type
				° C		° F			
				Min	Max	Min	Max		
Backhoe Loaders	All Points	High	Production work with long cycles and/or constant flow implements.	-35	40	-31	104	1	Ultra 5Moly Grease
				-30	50	-22	122	2	
		Medium	General work with regular cycles in medium applications.	-20	40	-4	104	2	Advanced 3Moly Grease
		Low	Utility work with intermittent cycles in light to medium applications.	-30	40	-22	104	2	

Diesel Fuel Recommendations

Diesel fuel must meet Caterpillar Specification for Distillate Fuel and the latest revisions of "ASTM D975-09a" and "EN 590" in order to ensure optimum engine performance. Refer to Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations" for the latest fuel information and for Cat fuel specification. This manual may be found on the Web at Safety.Cat.com.

Misfueling with fuels of higher sulfur level can have the following negative effects:

- Reduce engine efficiency and durability.

- Increase the wear.
- Increase the corrosion.
- Increase the deposits.
- Lower fuel economy
- Shorten the time period between oil drain intervals (more frequent oil drain intervals).
- Increase overall operating costs.

Failures that result for the use of improper fuels are not Caterpillar factory defects. Therefore, the cost of repairs would not be covered by a Caterpillar warranty.

Caterpillar does not require the use of ULSD in non-road and machine applications that are not Tier 4/Stage IIIB/ Stage IV certified engines and are not equipped with aftertreatment devices. For Tier 4/Stage IIIB/Stage IV certified engines, always follow operating instructions and fuel tank inlet labels to insure the correct fuels are used.

Refer to Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations" for more details about fuels, lubricants, and Tier 4 requirements. This manual may be found on the Web at Safety.Cat.com.

Fuel Additives

Cat Diesel Fuel Conditioner and Cat Fuel System Cleaner are available for use when needed. These products are applicable to diesel and biodiesel fuels. Consult your Cat dealer for availability.

Biodiesel

Biodiesel is a fuel that can be made from various renewable resources that include vegetable oils, animal fat, and waste cooking oil. Soybean oil and rapeseed oil are the primary vegetable oil sources. In order to use any of these oils or fats as fuel, the oils or fats are chemically processed (esterified). The water and contaminants are removed.

U.S. distillate diesel fuel specification "ASTM D975-09a" includes up to B5 (5 percent) biodiesel. Currently, any diesel fuel in the U.S. may contain up to B5 biodiesel fuel.

European distillate diesel fuel specification "EN 590" includes up to B5 (5 percent) and in some regions up to B7 (7 percent) biodiesel. Any diesel fuel in Europe may contain up to B5 or in some regions up to B7 biodiesel fuel.

Note: The diesel portion used in the biodiesel blend must be Ultra Low Sulfur Diesel (15 ppm sulfur or less, per "ASTM D975"). In Europe the diesel fuel portion used in the biodiesel blend must be sulfur free diesel (10 ppm sulfur or less, per "EN 590"). The final blend must have 15 ppm sulfur or less.

When biodiesel fuel is used, certain guidelines must be followed. Biodiesel fuel can influence the engine oil, aftertreatment devices, non-metallic, fuel system components, and others. Biodiesel fuel has limited storage life and has limited oxidation stability. Follow the guidelines and requirements for engines that are seasonally operated.

In order to reduce the risks associated with the use of biodiesel, the final biodiesel blend and the biodiesel fuel used must meet specific blending requirements.

All the guidelines and requirements are provided in the latest revision of Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations". This manual may be found on the Web at Safety.Cat.com.

Coolant Information

The information provided in this "Coolant Recommendation" section should be used with the "Lubricants Information" provided in the latest revision of Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations". This manual may be found on the Web at Safety.Cat.com.

The following two types of coolants may be used in Cat diesel engines:

Preferred - Cat ELC (Extended Life Coolant)

Acceptable - Cat DEAC (Diesel Engine Antifreeze/Coolant)

NOTICE

Never use water alone as a coolant. Water alone is corrosive at engine operating temperatures. In addition, water alone does not provide adequate protection against boiling or freezing.
